

YENEPOYA INSTITUTE OF ARTS, COMMERCE AND SCIENCE

A Constituent Unit of Yenepoya (Deemed to be University), Balmatta, Mangalore

Department of Computer Science

Project – High Level Design

on

Project Title

Data-Driven Risk Profiling for Supply Chain

Student Name: Mohammad Fazil Anwaz

Industry Mentor:

Guided by:

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1. Introduction

This document presents the High Level Design (HLD) for "Data-Driven Risk Profiling for Supply Chain" (PRJN26-250). It outlines system architecture, component interactions, and data flows for a weighted cyber risk scoring engine built using Python, Pandas, Matplotlib, and Seaborn in Jupyter Notebook.

The scoring engine enables analysts to systematically assess and rank supplier cybersecurity risk using industry frameworks including NIST CSF, ISO 27001, and CIS Controls v8.

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1.1 Scope of the Document

- **Overall system architecture and component breakdown**
- **Application design and technology stack**
- **Process flow and data flow design**
- **Data model and access mechanisms**
- **Non-functional requirements including security and performance**

1.2 Intended Audience

- Students — to understand system design before implementation
- Faculty / Guided by — to review architectural decisions
- Industry Mentors — to validate design against professional standards
- Evaluators — to assess project design completeness

1.3 System Overview

The system takes supplier-level input scores across six cybersecurity risk dimensions, applies a user-defined weighted scoring formula, and produces ranked risk classification with visual dashboards.

Risk Score = $\Sigma (\text{Weight}_i \times \text{Score}_i)$ for $i = 1$ to 6

2. System Design

The system is structured into three distinct layers:

<u>Layer</u>	<u>Responsibility</u>	<u>Technology</u>
<u>Input Layer</u>	<u>Load and validate supplier risk data</u>	<u>Pandas, CSV</u>
<u>Processing Layer</u>	<u>Apply weighted scoring formula and classify risk</u>	<u>Python, NumPy</u>
<u>Output Layer</u>	<u>Generate visualizations and export results</u>	<u>Matplotlib, Seaborn</u>

2.1 Application Design

Implemented as a Jupyter Notebook (.ipynb) with 5 modular cells:

- **Cell 1 — Library imports and environment setup**
- **Cell 2 — Risk factor definition, weight assignment, and dataset creation**
- **Cell 3 — Weighted risk score calculation and risk classification**
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- **Cell 4 — Visualization dashboard generation (bar charts, pie charts)**
- **Cell 5 — Results export and final summary report**

2.2 Process Flow

PROCESS FLOW

Step 1: Define Risk Factors & Assign Weights (sum = 1.0)



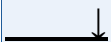
Step 2: Input Supplier Scores (1-10 scale per factor)



Step 3: Calculate Risk Score = $\Sigma (\text{Weight}_i \times \text{Score}_i)$



Step 4: Classify Risk — Low / Medium / High / Critical



Step 5: Generate Visual Dashboard & Export CSV Report **Figure 1: System Architecture Diagram**

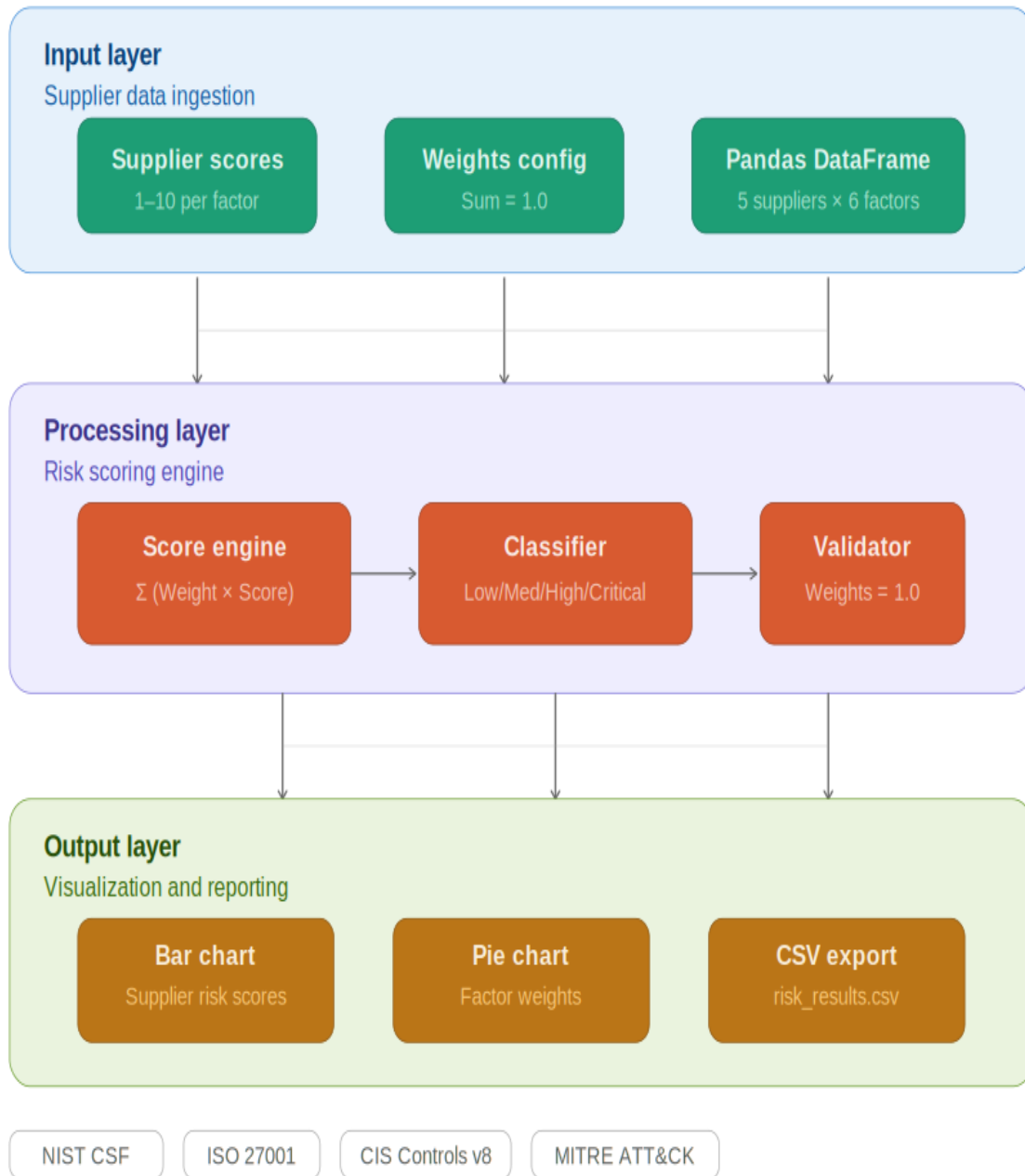
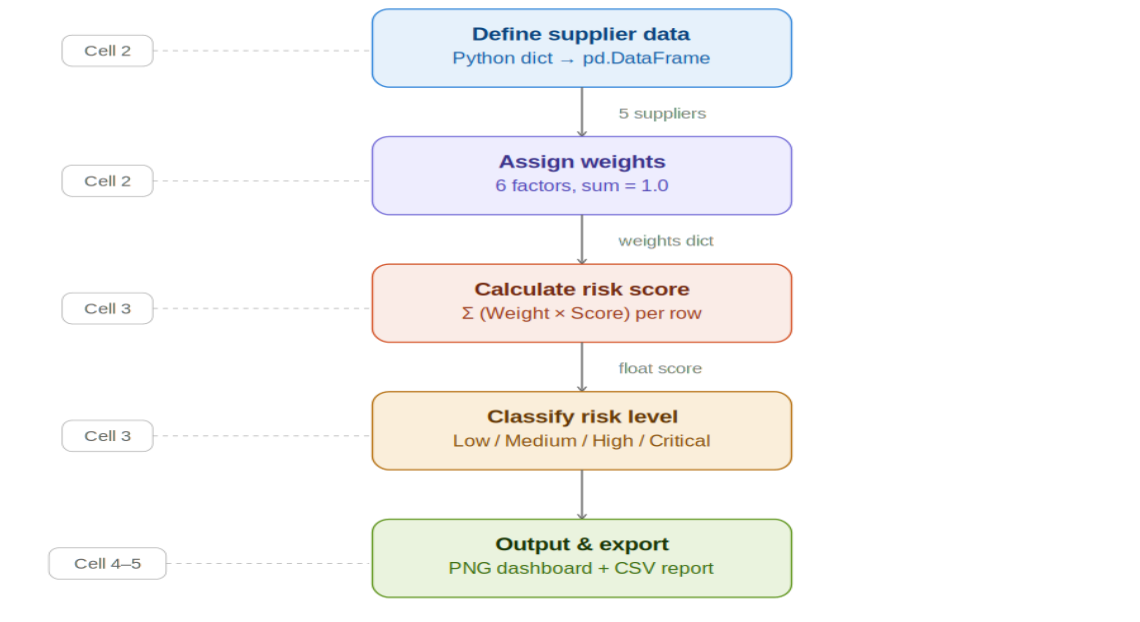


Figure 2: Data Flow Diagram

2.3 Information Flow

- **Raw supplier scores loaded into Pandas DataFrame**
- **Risk engine iterates over each row, applies weighted formula**
- **Calculated scores appended as new columns to DataFrame**
- **Classification module maps scores to risk tiers**
- **Output module renders charts and exports enriched DataFrame to CSV**

2.4 Components Design

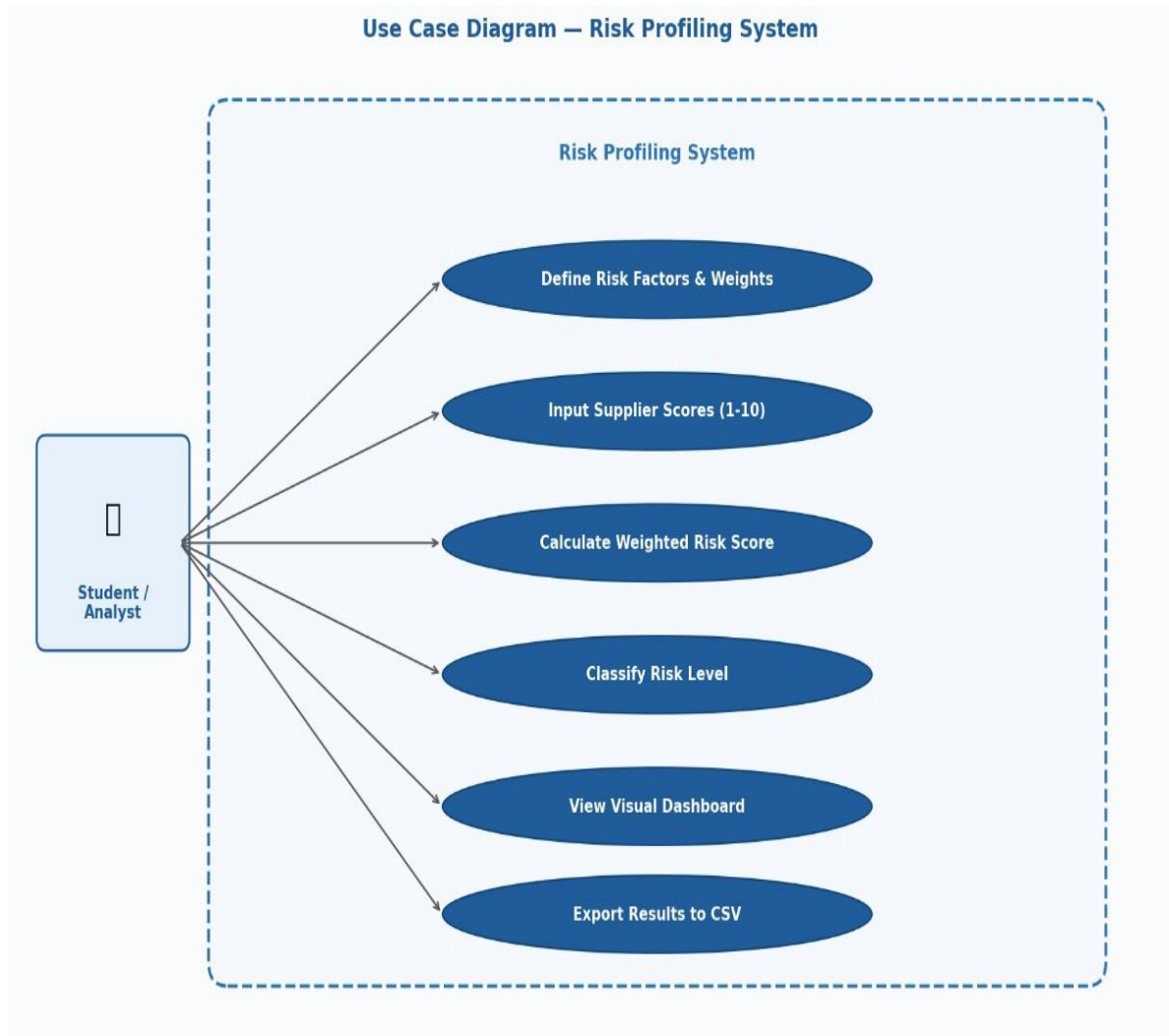
<u>Component</u>	<u>Function</u>	<u>Implementation</u>
<u>Data Input Module</u>	<u>Loads supplier data into DataFrame</u>	<u>pandas.DataFrame</u>
<u>Risk Calculation Engine</u>	<u>Applies weighted scoring formula</u>	<u>Python arithmetic</u>
<u>Classification Module</u>	<u>Maps scores to risk tiers</u>	<u>Python function</u>
<u>Visualization Module</u>	<u>Generates bar/pie charts</u>	<u>Matplotlib, Seaborn</u>
<u>Reporting Module</u>	<u>Exports results to CSV</u>	<u>pandas.to_csv()</u>

2.5 Key Design Considerations

- Transparency — fully interpretable weighted scoring model
- Flexibility — weights adjustable to reflect different risk priorities
- Scalability — additional suppliers or factors added without changing core logic
- Modularity — each notebook cell is independent and extendable
- Open Source — all libraries freely available with no licensing cost

2.6 Use Case Diagram

- **The Use Case Diagram below illustrates the interactions between the Student/Analyst actor and all system use cases within the Risk Profiling System boundary:**



- **Figure 3: Use Case Diagram — System Use Cases and Actor Interactions**

2.7 Sequence Diagram

The Sequence Diagram shows the chronological message flow between the student, Jupyter Notebook, and internal system modules during a complete risk profiling session:

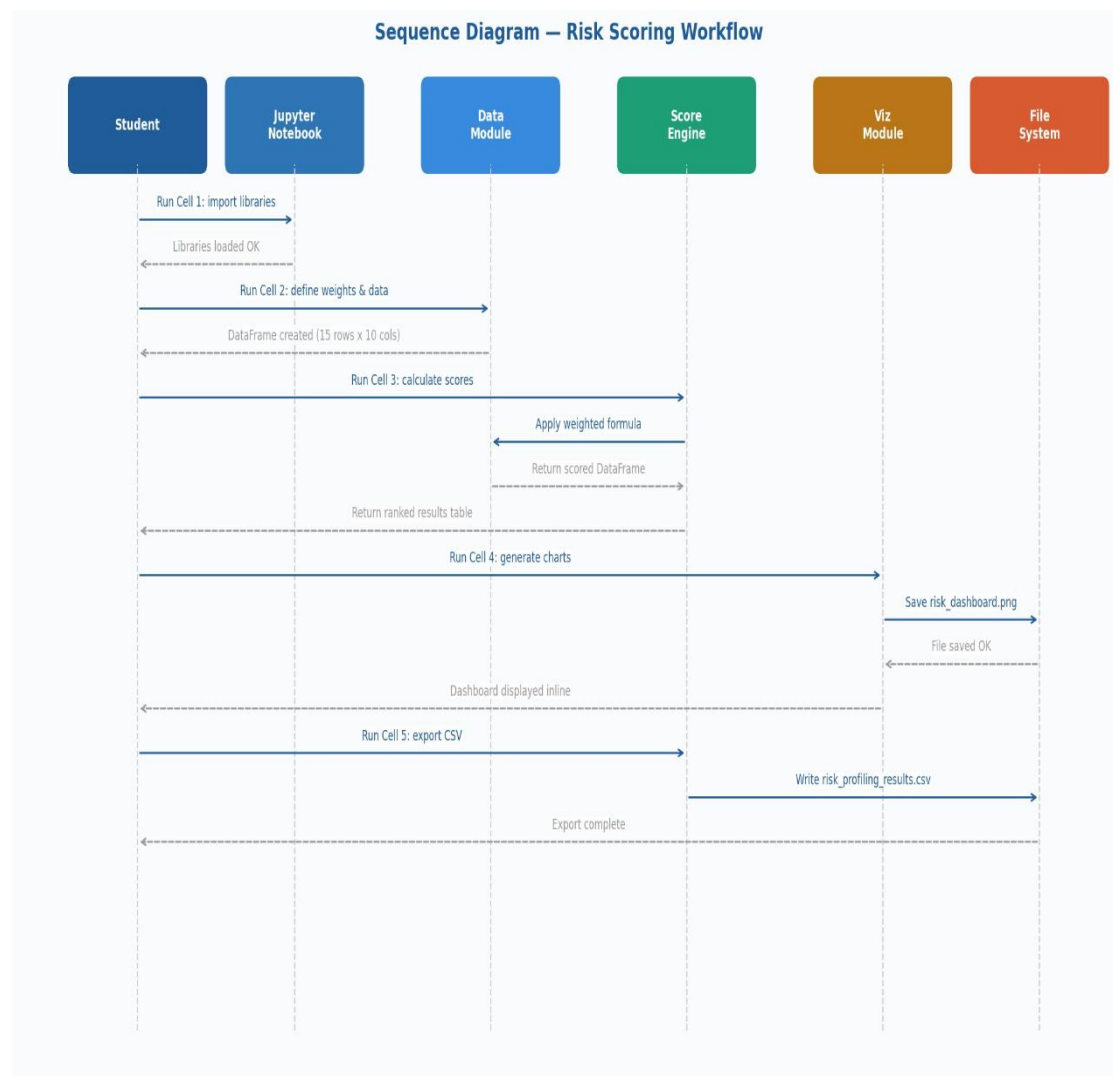


Figure 4: Sequence Diagram — Risk Scoring Workflow Message Flow

2.8 Component Diagram

The Component Diagram illustrates all system components, their individual responsibilities, inter-component relationships, and open-source library dependencies:

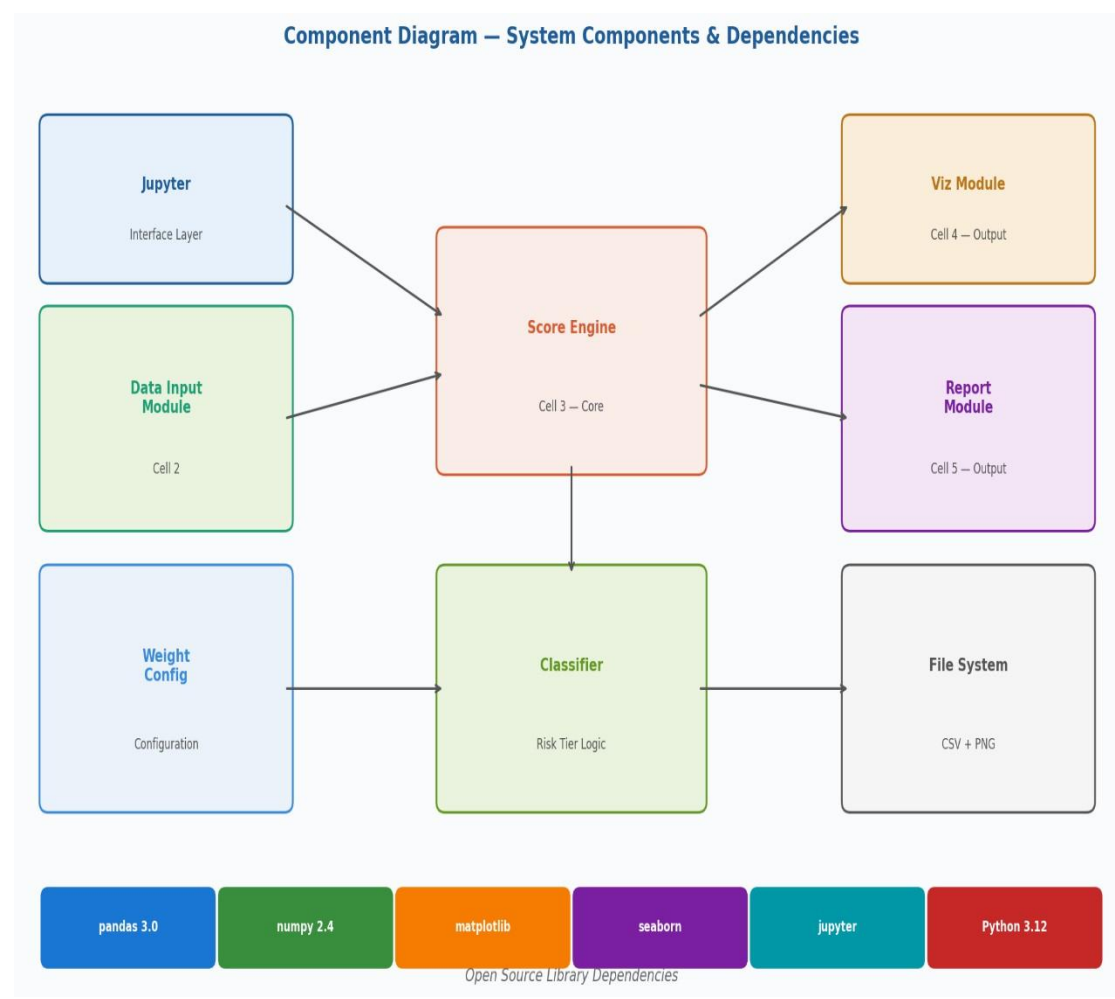


Figure 5: Component Diagram — System Components and Dependencies

2.9 Activity Flow Diagram

The Activity Flow Diagram shows the complete step-by-step workflow including decision points for library validation and weight verification:

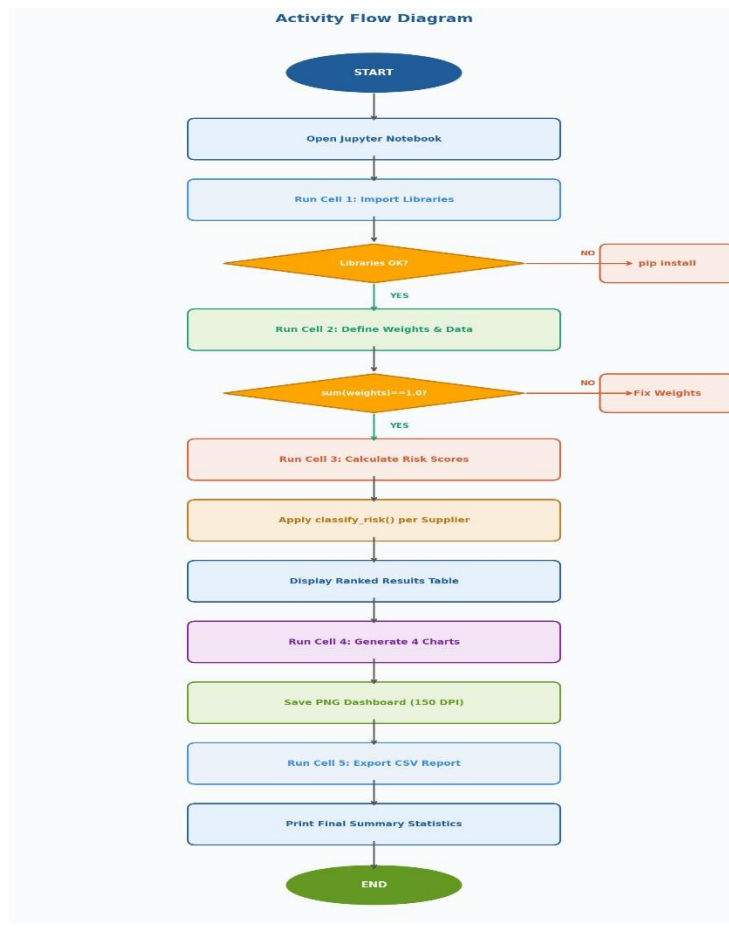


Figure 6: Activity Flow Diagram — Complete Workflow with Decision Poin

3. Data Design

3.1 Data Model

<u>Column Name</u>	<u>Data Type</u>	<u>Range</u>	<u>Description</u>
<u>Supplier</u>	<u>String</u>	<u>N/A</u>	<u>Name of the supplier</u>
<u>Vendor Financial Health</u>	<u>Integer</u>	<u>1 – 10</u>	<u>Financial stability score</u>
<u>Third Party Access Control</u>	<u>Integer</u>	<u>1 – 10</u>	<u>Access control score</u>
<u>Incident Response Readiness</u>	<u>Integer</u>	<u>1 – 10</u>	<u>IR readiness score</u>
<u>Vulnerability Management</u>	<u>Integer</u>	<u>1 – 10</u>	<u>Vulnerability mgmt score</u>
<u>Delivery SLA Performance</u>	<u>Integer</u>	<u>1 – 10</u>	<u>SLA performance score</u>
<u>Compliance Certification</u>	<u>Integer</u>	<u>1 – 10</u>	<u>Compliance score</u>
<u>Risk Score</u>	<u>Float</u>	<u>0.0 – 10.0</u>	<u>Calculated weighted score</u>
<u>Risk Level</u>	<u>String</u>	<u>N/A</u>	<u>Low/Medium/High/Critical</u>

3.2 Data Access Mechanism

- **Supplier data defined as Python dictionary, converted to Pandas DataFrame**
- **Factor scores accessed using column indexing: df['column_name']**
- **Weighted scores calculated using vectorized Pandas operations**
- **Final results exported using df.to_csv() for persistent storage**

3.3 Data Retention Policies

- **All data stored locally — no cloud storage used**
- **Results exported to risk_profiling_results.csv in project directory**
- **Visualizations saved as risk_dashboard.png for documentation**
- **No personally identifiable information (PII) stored or processed**

4. Interfaces

<u>Interface</u>	<u>Type</u>	<u>Description</u>
<u>Jupyter Notebook</u>	<u>Web-based GUI</u>	<u>Primary interface for code execution</u>
<u>CSV Output</u>	<u>File Interface</u>	<u>Exported results for reporting</u>
<u>PNG Dashboard</u>	<u>Image Output</u>	<u>Visual risk dashboard chart</u>
<u>Terminal / VS Code</u>	<u>CLI</u>	<u>Used to launch Jupyter server</u>

5. Non-Functional Requirements

5.1 Security Aspects

- **System runs entirely on local machine — no network connectivity required**
- **No sensitive or real supplier data used — all data is synthetic**
- **Output files stored locally and do not leave the user's environment**

5.2 Performance Aspects

- **Processes up to 100 suppliers instantly using vectorized Pandas operations**
- **Chart generation via Matplotlib completes within 2-3 seconds**
- **CSV export is near-instantaneous for datasets up to 10,000 rows**

6. References

- **NIST Cybersecurity Framework (CSF) v1.1 — <https://www.nist.gov/cyberframework>**
- **ISO/IEC 27001:2022 — Information Security Management Systems**
- **CIS Controls v8 — <https://www.cisecurity.org>**
- **MITRE ATT&CK for Supply Chain — <https://attack.mitre.org>**
- **Python 3.12 Documentation — <https://docs.python.org/3.12>**
- **Pandas Documentation — <https://pandas.pydata.org/docs>**
- **Matplotlib — <https://matplotlib.org>**
- **Jupyter Notebook — <https://jupyter.org>**